

LESSON 5-5 • SESSION 2 • PREREQUISITE PRACTICE B • SKILL 1 OF 2

Volume of Rectangular Prisms

Builds on Lesson 5-5 Session 1 • 6.G.A.2

TODAY'S LESSONYou will use $V = l \times w \times h$ with fractions, like $22 \times 9 \times \frac{1}{2}$ for a slab.**YOU NEED TO**Find volume: base area $\times$ height, like $15 \times 8 \times 6$. **You will use it in Practice Form A, problems 1 and 6.****STEPS TO FOLLOW***use these same steps on every problem*

- 1 Find** the base area: length $\times$ width.
- 2 Multiply** the base area by the height.
- 3 Write** cubic units, like in^3 .

WATCH ME*the steps in action***A bin is 15 in long, 8 in wide, and 6 in tall. Find its volume.****Step 1** $B = 15 \times 8 = 120$.**Step 2** $V = 120 \times 6 = 720$.**Step 3** Unit: in^3 .**Answer** **720 in^3** **TRY IT WITH HELP***fill in each step*

- 1.** A box is 10 cm by 7 cm by 4 cm. Find its volume.

Step 1 $B = 10 \times 7 =$ _____**Step 2** $V = 70 \times 4 =$ _____**Step 3** Unit: _____

- 2.** A box is 9 ft by 4 ft by 5 ft. Find its volume.

Step 1 $B = 9 \times 4 =$ _____**Step 2** $V = 36 \times 5 =$ _____**Step 3** Unit: _____**YOUR TURN***use the steps on your own*

- 3.** Find each volume.

a. $5 \text{ m} \times 4 \text{ m} \times 3 \text{ m}$ _____**b.** $6 \text{ ft} \times 6 \text{ ft} \times 2 \text{ ft}$ _____**c.** $8 \text{ cm} \times 5 \text{ cm} \times 5 \text{ cm}$ _____**d.** $12 \text{ in} \times 3 \text{ in} \times 2 \text{ in}$ _____**e.** $11 \text{ m} \times 10 \text{ m} \times 3 \text{ m}$ _____**f.** $7 \text{ cm} \times 7 \text{ cm} \times 7 \text{ cm}$ _____

Find a Missing Dimension

Builds on Lesson 5-2 Session 2 • 6.G.A.1

TODAY'S LESSONYou will find a missing edge, like the width of a 5,400 in³ tank.**YOU NEED TO**Find a missing side: total $\div$ the side you know. **You will use it in Practice Form A, problems 2, 3, and 7.****STEPS TO FOLLOW***use these same steps on every problem*

- 1 Write** the fact with a ? for the missing number.
- 2 Divide** the total by the number you know.
- 3 Check:** multiply back to get the total.

WATCH ME*the steps in action***A rectangle has an area of 252 m². Its width is 12 m. Find its length.****Step 1** $? \times 12 = 252$.**Step 2** $252 \div 12 = 21$.**Step 3** $21 \times 12 = 252$. It checks.**Answer** **21 m****TRY IT WITH HELP***fill in each step*

- 4.**
- A rectangle has an area of 96 cm
- ²
- . One side is 8 cm. Find the other side.

Step 1 $? \times 8 = \underline{\hspace{2cm}}$ **Step 2** $96 \div 8 = \underline{\hspace{2cm}}$ **Step 3** $12 \times 8 = \underline{\hspace{2cm}}$

- 5.**
- A rectangle has an area of 130 in
- ²
- . One side is 13 in. Find the other side.

Step 1 $13 \times ? = \underline{\hspace{2cm}}$ **Step 2** $130 \div 13 = \underline{\hspace{2cm}}$ **Step 3** $10 \times 13 = \underline{\hspace{2cm}}$ **YOUR TURN***use the steps on your own*

- 6.**
- Find each missing number ?.

a. $? \times 9 = 117$ ____

b. $20 \times ? = 340$ ____

c. $? \times 15 = 225$ ____

d. $11 \times ? = 198$ ____

e. $? \times 25 = 400$ ____

f. $30 \times ? = 540$ ____

READY CHECK*a problem like the one in today's lesson*

- 7.**
- A box holds 600 in
- ³
- . Its base area is 50 in
- ²
- . How tall is it?

A 550 in**B** 12 in**C** 650 in**D** 30,000 in**CHECK YOURSELF**☐ I can do this on my own☐ I need a little help☐ I have a question about #____